

Downstream-Utility-Aware Source-Frame Selection for Reference-Based Video Color Grading

Yang Han^{1,2*}, Yaosen Chen^{1,2}, Yanru Zhang¹, Xin Duan²

^{1*}University of Electronic Science and Technology of China, Chengdu, 611731, China.

²Chengdu Sobey Digital Technology Co., Ltd., Chengdu, 610041, China.

*Corresponding author(s). E-mail(s): yanghan@sobey.com;

Abstract

Reference-based video color grading estimates one global color transform from a target image and one selected frame of a source video. Existing systems commonly choose that frame by global image similarity, although the most similar frame may not contain the structural correspondences and local color evidence needed to condition a transform for the entire sequence. We formulate source-frame selection as a downstream decision and introduce a three-stage selector that establishes semantic correspondence, compares local colors at corresponding regions, and performs set-aware utility ranking. Utilities are generated offline from training renders; at inference the selector uses only the reference and candidate frames, while the grading backend remains unchanged. On a 100-case benchmark, the complete system reaches **27.43** dB PSNR versus the **24.55** dB reported for VCG, an arithmetic difference of **2.88** dB. In a matched study that changes only the selection rule, the selector lowers perceptual and color-difference error. The results show that optimizing the conditioning observation is an effective complement to improving the downstream transform model.

Keywords: source-frame selection, reference-based video color grading, 3D look-up table, utility learning, structural correspondence, color matching

1 Introduction

Reference-based video color grading transfers the appearance of a target image to a source video. In a typical workflow, an artist or an automated system chooses one representative source frame, estimates a global color transform (often a three-dimensional look-up table, or 3D LUT) from that frame and the reference, and applies the transform to the complete sequence. This one-transform design is efficient and temporally stable, but it makes the choice of the conditioning frame consequential: every subsequent frame is graded using evidence gathered from that single observation.

The source observation is therefore an upstream decision problem, not merely a retrieval step. A video offers many candidate frames that differ in pose, object visibility, illumination, and color distribution. Figure 1 shows the practical ambiguity: candidates from the same shot can depict the same subject while providing different visible regions and color statistics. A frame that is close in a global embedding can still be a poor measurement for the transform, whereas a less similar frame can expose the correspondences that make the transform useful for the whole video.

Most existing systems treat this decision as image retrieval and choose the candidate with the

Why semantic similarity is not enough for source-frame selection

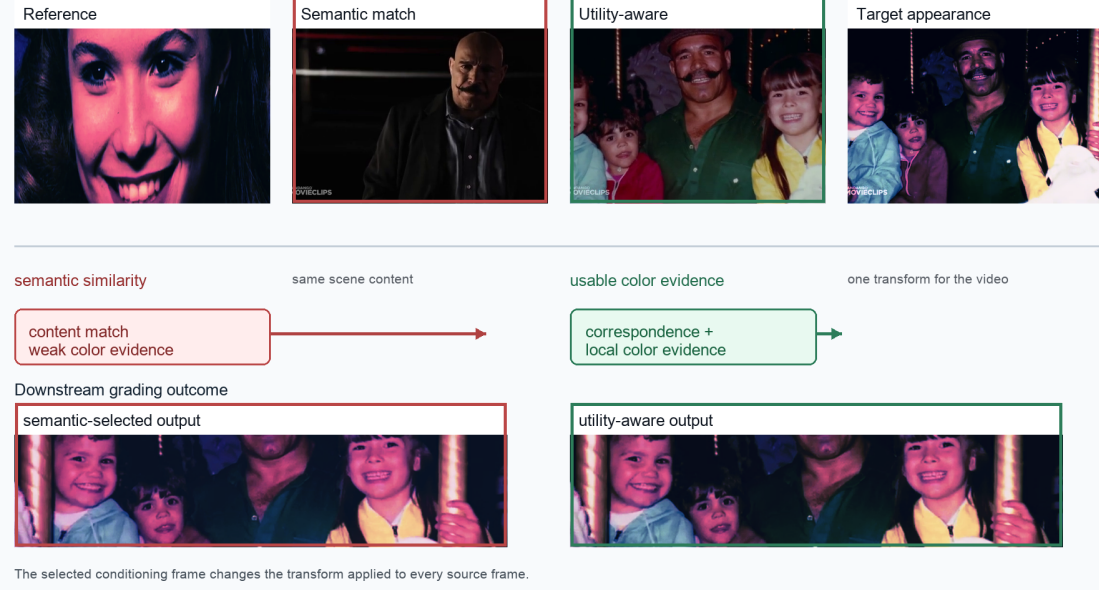


Fig. 1 A representative failure of semantic-only source-frame selection. The semantic-match candidate depicts related content but provides a weak chromatic condition for the downstream transform. The utility-aware choice supplies a more useful color observation, leading to a different graded video frame. All panels are rendered from one frozen evaluation case; the target frame is used only for offline evaluation and visualization.

highest global embedding similarity to the reference. That criterion is useful for finding related content, but it is not the objective of transform estimation. A useful conditioning frame must expose corresponding structures, carry informative local color relationships, and be representative of the source video rather than an atypical moment. These requirements are coupled and cannot be checked reliably by a single global similarity value.

We address this mismatch by formulating source-frame selection as downstream utility learning. Given a reference r and candidate set $\mathcal{C} = \{c_i\}_{i=1}^N$, a fixed grading process produces a measurable video-level utility q_i when conditioned on candidate c_i . The proposed selector learns candidate scores

$$f_{\theta}(r, \mathcal{C}) = (s_1, \dots, s_N), \quad i^* = \arg \max_i s_i, \quad (1)$$

so that the selected frame is optimized for the quality of the resulting video, rather than for an intermediate similarity measure. Utilities are generated only from the training partition; inference requires neither a target video nor ground-truth quality values.

The selector follows three explicit design principles. First, *semantic correspondence* identifies which candidate regions can be associated with the reference. Second, *correspondence-guided color comparison* measures chromatic relations at those associated regions rather than at fixed image coordinates. Third, *set-aware utility ranking* compares all candidates jointly before making the decision. The stages answer three different questions—where should evidence be compared, what evidence is useful, and which candidate is best among the available alternatives—which makes the architecture auditable and preserves the one-frame interface of existing grading systems.

We instantiate the formulation with VCG only for quantitative validation and leave that grading backend unchanged. The complete system reaches 27.43 dB PSNR on the 100-case benchmark; a matched experiment that changes only the selection rule also improves perceptual and color-difference measures. Temporal, semantic, chromatic, oracle, and simpler learned controls then locate the method’s benefit and its remaining headroom.

The main contributions are as follows:

- We identify the conditioning frame as an optimization variable in reference-based video color grading and formulate its selection through video-level downstream utility rather than generic image similarity.
- We introduce a three-stage selector that integrates semantic correspondence, correspondence-guided local color comparison, and set-based utility ranking while preserving the standard one-frame interface of existing grading systems.
- We provide a controlled evaluation that separates complete-system performance from the selector’s contribution and reports perceptual, colorimetric, stability, and case-level evidence.

2 Related Work

2.1 Reference-based color transformation

Reference-based grading estimates a color transformation that transfers a target appearance while preserving source content. Global 3D LUTs are widely used because they provide expressive non-linear color mappings with inexpensive trilinear evaluation. Learning-based methods have improved LUT generation through image-adaptive basis mixtures [1], adaptive sampling intervals [2], factorized representations [3], and compact parameterizations [4]. Video-oriented systems extend these ideas with keyframe adaptation and temporal application [5, 6].

This literature primarily studies how to estimate or represent the color operator. Our work addresses an upstream variable: which source observation should condition that estimation. The distinction matters whenever a global operator is inferred from a reference and only one or a small number of source frames.

2.2 Frame selection and task utility

Classical keyframe selection targets summarization, coverage, or diversity so that a small set of frames remains representative of a video [7]. Reference-based grading has a different objective. A representative or semantically similar frame can still be poorly conditioned for estimating a color transform. VCG makes the decision explicit but selects the source frame by CLIP similarity [6]. We instead supervise the decision with the quality

obtained after the complete downstream grading operation.

This formulation is related to learning to rank. Pairwise methods learn local preference relations [8], whereas listwise methods match a distribution over a complete candidate set [9]. Our principal objective retains the measured magnitude of within-video utility differences through centered robust regression. Pairwise and listwise losses are evaluated as alternatives rather than assumed to be the natural objective for grading.

2.3 Structure and appearance correspondence

Reliable comparison of a source frame and a target reference requires both structural association and chromatic evidence. Self-supervised patch features, including DINOv2 descriptors, provide strong spatial correspondences across changes in pose and appearance [10]. Global image embeddings such as CLIP summarize high-level image context [11], but do not preserve the local color relationships required by transform estimation.

The separation between structure and appearance also appears in neural style transfer. Content activations and feature statistics are treated as complementary signals [12, 13], and photorealistic methods add spatial constraints to prevent invalid correspondence [14]. Our method adapts this principle to a decision problem: patch descriptors determine where candidate and reference regions correspond, while local RGB tokens determine how their colors differ. A candidate-set Transformer without positional embeddings then compares the available frames, following the general principle of permutation-equivariant set processing [15].

In contrast to methods that synthesize or regularize video frames [16–18], the proposed module does not alter the rendering process or introduce a temporal loss. It selects the conditioning observation from which an existing grading system estimates its global transform.

3 Method

3.1 Problem formulation and overview

Let $V = \{v_t\}_{t=1}^T$ denote a source video, r a target reference image, and $\mathcal{C} = \{c_i\}_{i=1}^N$ a set of candidate

frames sampled from V . A reference-based grading operator $\mathcal{G}$ estimates a global transform from a candidate-reference pair and applies it to the video:

$$L_i = \mathcal{G}(c_i, r), \quad \hat{V}_i = L_i(V). \quad (2)$$

Different candidates produce different transforms and therefore different graded videos. For training cases with target video Y , we define the utility of candidate c_i as

$$q_i = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathcal{Q}(\hat{y}_{i,t}, y_t), \quad (3)$$

where $\mathcal{Q}$ is a task-quality measure and $\mathcal{T}$ is a set of temporal samples. This computation is performed offline and only on the training partition. The selector learns a score s_i for every candidate and returns

$$i^* = \arg \max_{i \in \{1, \dots, N\}} s_i. \quad (4)$$

At inference, neither Y nor q_i is available; the selected frame is passed to the otherwise unchanged grading operator.

The method is organized into the three stages shown in Fig. 2. Structural correspondence identifies related regions across the candidate and reference. The resulting association guides local color matching. These local relations are combined with global appearance context and compared across the candidate set to predict relative grading utility.

3.2 Structural correspondence

A frozen self-supervised image encoder extracts patch descriptors from each candidate and the reference. After normalization, spatial pooling, and a trainable projection, the descriptors are represented as

$$\begin{aligned} S_i &= \phi_s(\mathcal{D}(c_i)), & S_r &= \phi_s(\mathcal{D}(r)), \\ S_i, S_r &\in \mathbb{R}^{P \times d_s}, \end{aligned} \quad (5)$$

where P is the number of spatial tokens. Multi-head cross-attention uses the candidate as query and the reference as key and value. For head h ,

$$A_i^{(h)} = \text{softmax} \left(\frac{(S_i W_Q^{(h)})(S_r W_K^{(h)})^\top}{\sqrt{d_h}} \right), \quad (6)$$

and the structural correspondence map is

$$A_i = \frac{1}{H} \sum_{h=1}^H A_i^{(h)}, \quad \tilde{S}_i = \text{MHA}(S_i, S_r, S_r). \quad (7)$$

This soft association is preferable to comparing fixed coordinates because the candidate and reference may differ in pose, framing, or scene layout.

3.3 Correspondence-guided local color matching

The structural encoder is designed for visual correspondence rather than precise chromatic measurement. We therefore preserve local RGB information in a separate trainable convolutional stream. Let

$$C_i = \phi_c(c_i), \quad C_r = \phi_c(r), \quad C_i, C_r \in \mathbb{R}^{P \times d_c} \quad (8)$$

denote local color tokens on the same spatial grid as the structural tokens. Instead of estimating another unconstrained correspondence, the structural map aligns the reference color tokens to the candidate:

$$\tilde{C}_i = A_i C_r. \quad (9)$$

The pair $(C_i, \tilde{C}_i)$ describes how candidate colors relate to the reference at structurally associated regions. This design decouples the roles of the two representations: learned patch descriptors establish where to compare, while RGB features retain the color evidence used in that comparison.

3.4 Set-aware utility ranking

Local correspondence alone does not capture the overall appearance of an image. A frozen image encoder therefore provides normalized global descriptors g_i and g_r , which are mapped to a common appearance space by ϕ_g . For candidate token p , structural, chromatic, and global pair relations are formed with the standard comparison operators

$$\rho(x, y) = [x, y, x - y, x \odot y]. \quad (10)$$

The fused relation token is

$$R_{i,p} = \psi([\rho(S_{i,p}, \tilde{S}_{i,p}), \rho(C_{i,p}, \tilde{C}_{i,p}), \rho(\phi_g(g_i), \phi_g(g_r))]). \quad (11)$$

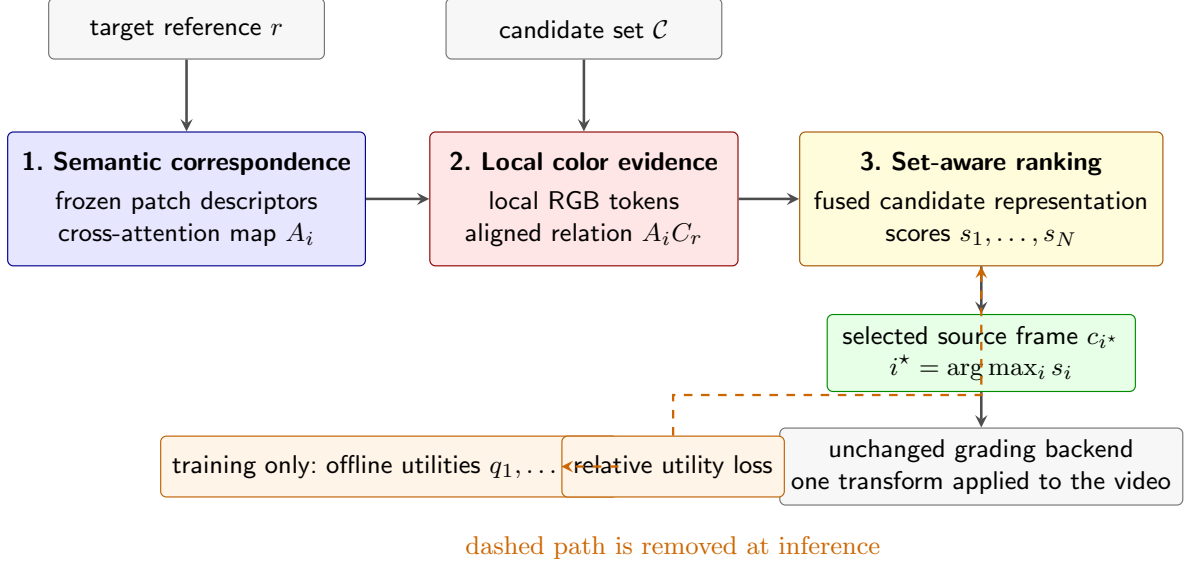


Fig. 2 Overview of the proposed source-frame selector. The three numbered stages form the technical core: patch-level structural correspondence, correspondence-guided local color matching, and set-aware utility ranking. Offline grading utilities supervise the predicted scores during training only; inference ends with a single selected source frame.

where ψ is a trainable multilayer perceptron. Attention pooling converts the P relation tokens into one candidate representation:

$$h_i^{(0)} = \phi_p(\text{vec}(\text{MHA}(Q_{\text{pool}}, R_i, R_i))). \quad (12)$$

The candidates are then processed jointly by a Transformer encoder without positional embeddings:

$$[h_1, \dots, h_N] = \text{SetEnc}([h_1^{(0)}, \dots, h_N^{(0)}]), \quad s_i = \phi_u(h_i). \quad (13)$$

Thus, the score of a frame can depend on the other available candidates. The absence of candidate-index embeddings preserves permutation equivariance. For any permutation matrix Π and score function F_θ ,

$$F_\theta(\Pi H^{(0)}) = \Pi F_\theta(H^{(0)}), \quad \Pi \in \mathbb{R}^{N \times N}. \quad (14)$$

3.5 Relative utility learning

Absolute grading quality varies across videos and target appearances. The selection decision, however, depends on differences among candidates within the same case. We therefore center the

offline utilities and predicted scores:

$$\tilde{q}_{b,i} = q_{b,i} - \frac{1}{N} \sum_{j=1}^N q_{b,j}, \quad \tilde{s}_{b,i} = s_{b,i} - \frac{1}{N} \sum_{j=1}^N s_{b,j}. \quad (15)$$

The principal objective is robust regression of this relative utility:

$$\mathcal{L}_{\text{util}} = \frac{1}{BN} \sum_{b=1}^B \sum_{i=1}^N \text{SmoothL1}_\beta(\tilde{s}_{b,i} - \tilde{q}_{b,i}), \quad (16)$$

where B is the batch size and β is the transition parameter. Centering removes case-level quality offsets while preserving both candidate ordering and the magnitude of useful within-case differences.

For diagnostic comparisons, we also evaluate standard pairwise and listwise ranking objectives. The pairwise alternative applies logistic ranking to ordered candidate pairs $\mathcal{P}_b$:

$$\mathcal{L}_{\text{pair}} = \frac{1}{\sum_b |\mathcal{P}_b|} \sum_b \sum_{(i,j) \in \mathcal{P}_b} \ell_{b,i,j}, \quad (17)$$

$$\ell_{b,i,j} = \log(1 + \exp[-(s_{b,i} - s_{b,j})]).$$

where $(i, j) \in \mathcal{P}_b$ only when candidate i has greater measured utility than candidate j by a prescribed margin. The listwise alternative matches the score

and utility distributions:

$$\mathcal{L}_{\text{list}} = -\frac{1}{B} \sum_{b=1}^B \sum_{i=1}^N p_{b,i} \log \hat{p}_{b,i},$$

$$p_{b,i} = \text{softmax}_i(q_{b,i}/\tau), \quad \hat{p}_{b,i} = \text{softmax}_i(s_{b,i}/\tau). \quad (18)$$

These objectives replace, rather than augment, Eq. (16) in the corresponding sensitivity experiments.

3.6 Training and inference

The structural and global image encoders remain frozen, and their features can be cached. The local color encoder, relation layers, attention pooling, candidate-set encoder, and utility head are optimized with the offline labels from Eq. (3). Candidate order is randomized during training. At inference, the selector receives only r and $\mathcal{C}$, predicts $\{s_i\}_{i=1}^N$, and returns the frame selected by Eq. (4). The grading operator, target transform, target video, and candidate utilities are not inputs to the selector.

4 Experiments

The evaluation is arranged as a chain of evidence. We first report the complete grading system on the benchmark. We then hold the grading backend and candidate pool fixed to isolate the contribution of the selection rule, compare against simple alternatives, and finally examine stability, candidate availability, selection behaviour, and visual output. This ordering keeps the principal claim separate from diagnostic analyses of how the selector makes its decision.

4.1 Data and evaluation protocol

The training partition contains 2,900 reference–video cases constructed from 87 available color–transform identities. The evaluation partition contains 100 cases from ten identities disjoint from training. Each case consists of a target reference image, a source video, and a common pool of N temporally sampled candidate frames. Every selector sees exactly this pool. After selection, the unchanged grading backend estimates one transform from the reference and the selected frame and applies it to the complete source video. Target frames, target transforms, and rendered utilities

are used only by offline label generation or evaluation; they are never inputs to the selector at inference.

For a frame-level metric m , we first average over the decoded frames of one case and then average the case means:

$$\bar{m}_c = \frac{1}{T} \sum_{t=1}^T m(\hat{y}_{c,t}, y_{c,t}), \quad \bar{m} = \frac{1}{M} \sum_{c=1}^M \bar{m}_c, \quad (19)$$

where T is the number of evaluated frames and M is the number of cases. PSNR is evaluated after clipping to the normalized sRGB range and converting to the implementation’s 8-bit representation. LPIPS uses its standard AlexNet backbone with inputs in $[-1, 1]$. CIEDE2000 is computed after sRGB-to-CIELAB conversion with a D65 white point and averaged over pixels. Confidence intervals are obtained by resampling transform identities and cases in a hierarchical bootstrap.

4.2 Implementation

The structural and global image encoders are frozen and their features are cached. All remaining selector layers are trained with AdamW (batch size 8, learning rate 5×10^{-4} , and gradient-norm clipping at 5) using the centered relative-utility objective in Eq. (16). The selector has 3.53 million trainable parameters. Its structural, local-color, global-appearance, and fused-relation widths are $d_s = 128$, $d_c = 32$, $d_g = 64$, and $d_r = 256$, respectively. Cross-attention uses four heads, relation pooling uses eight heads, and the candidate-set encoder uses three layers with eight heads and feed-forward width 1,024. Five transform-disjoint development folds determine a fixed seven-epoch schedule for the final fit. Training utilities use 60 temporal samples at the implementation resolution; the final evaluator uses all 480 decoded frames. Random initialization details, manifests, and checksums are supplied in the reproducibility archive.

4.3 Principal results

The first question is whether the complete pipeline improves benchmark grading quality. Table 1 answers this question with two clearly separated evidence sources. The upper panel is the complete-system benchmark result: the proposed selector

combined with the VCG grading backend reaches 27.43 dB. The 24.55 dB number is the value reported in the VCG paper under that paper’s protocol and is included only as contextual reference. The lower panel is the attribution experiment: the local renderer, candidate pool, rendering procedure, and evaluator are held fixed, and only the source-frame rule is changed. Under this matched protocol the proposed rule lowers LPIPS by 0.0043 and CIEDE2000 by 0.3782.

The matched metrics measure complementary failure modes: LPIPS captures learned perceptual discrepancy and CIEDE2000 captures color difference. Their favorable changes in the same controlled comparison show that the effect is not tied to one error definition. The corresponding paired intervals are reported in the supplementary results record.

4.4 Alternative selection rules and learned controls

We next ask whether the gain is specific to downstream utility learning and whether set-level comparison adds value beyond feature capacity. All rows in Table 2 use the common local evaluator. The linear regressor learns utility from frozen CLIP features, whereas the candidate-wise multimodal MLP scores each frame independently and therefore removes candidate-set interaction. The proposed model is reported with the same three training runs used for the stability analysis. These are simpler learned controls, not parameter-matched replicas.

Utility supervision gives a clear improvement over direct CLIP retrieval, even with a linear predictor. The set-aware selector has the highest mean among the learned controls, while the RGB mean-standard-deviation heuristic remains a competitive non-learning rule (27.5168 dB in the full baseline matrix). We therefore do not claim that the proposed architecture dominates every hand-crafted statistic; the controlled result establishes the benefit of the decision objective, and the controls locate the additional effect of joint candidate reasoning. The oracle score shows that the candidate pool retains substantial headroom.

4.5 Stability and candidate availability

Table 3 reports the three independent fits of the principal selector. The frozen principal run used for the benchmark headline is 27.4295 dB; the table reports the corresponding multi-run distribution for stability rather than replacing that headline with a different estimator.

Because the set encoder has no candidate-position embedding, the same interface supports different pool sizes. On uniform subsets of the temporal grid, PSNR increases from 27.1856 dB with five candidates to 27.4295 dB with twenty candidates (Table 4). This trend reflects access to more observations, not a change in the grading backend.

4.6 Utility and decision diagnostics

The offline training target is a 60-frame temporal proxy, whereas final evaluation averages over the complete decoded sequence. The proxy preserves the full-utility ordering under the fixed protocol: pooled Pearson and Spearman correlations are 0.99995, the mean within-case Spearman correlation is 0.99940, and the proxy and full-utility top-1 choices agree in 98% of cases. The proxy therefore reduces label-generation cost without changing the selected candidate in almost all cases.

The selector is a decision rule rather than a calibrated estimator of every candidate’s final utility. Its mean within-case Spearman correlation with the final utility is 0.0994, exact oracle top-1 accuracy is 9%, and the oracle lies in its top five in 32% of cases. These diagnostics expose the remaining selection uncertainty without introducing another aggregate performance claim.

4.7 Qualitative analysis

Figure 3 makes the selection problem visible. The reference and selected source frame need not have identical composition; what matters is that the selected observation provides usable structural and color evidence for estimating a transform that is then applied to another source frame. The examples cover cool, magenta, red, and warm looks. The residual maps show where the rendered result still differs from the target and complement the aggregate metric table.

Table 1 Principal evidence on the 100-case benchmark. The upper panel is a complete-system result and uses the value reported by the original VCG paper as an external contextual reference. The lower panel is a matched comparison in which only the source-frame selection rule changes.

Evidence	System or selection rule	PSNR $\uparrow$	LPIPS $\downarrow$	CIEDE2000 $\downarrow$
Complete system	VCG reported value [6]	24.55	—	—
	Proposed selector + VCG	27.43	—	—
	Arithmetic difference (context only)	+2.88	—	—
Matched selection study	VCG CLIP rule	—	0.0459	5.2014
	Proposed selector	—	0.0415	4.8232
	Difference (proposed – CLIP)	—	–0.0043	–0.3782

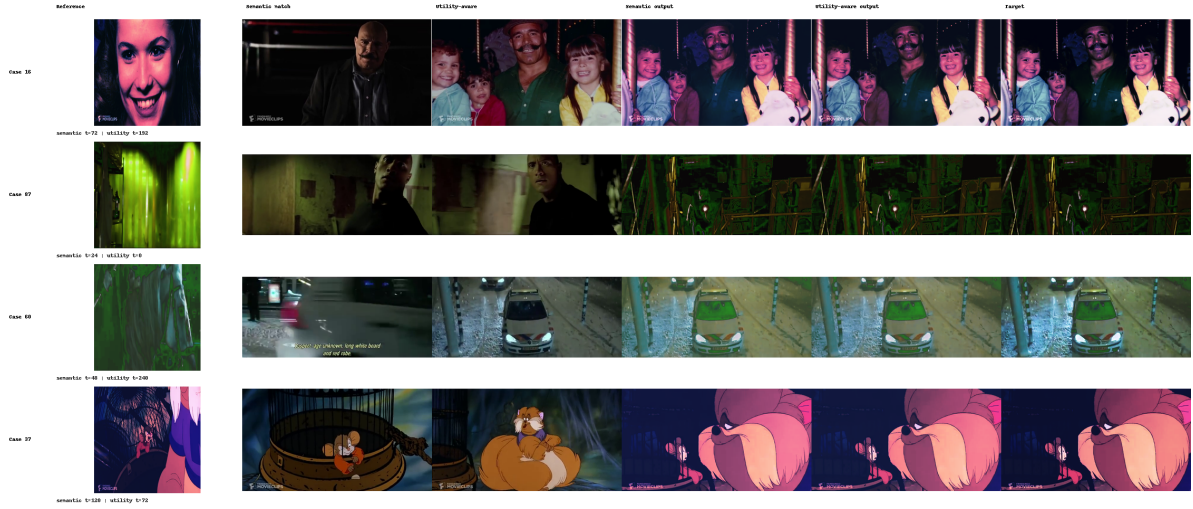


Fig. 3 Matched qualitative comparison on four frozen evaluation cases. Each row shows the target reference, the frame selected by semantic similarity, the frame selected by the proposed utility-aware rule, and the corresponding graded outputs, followed by the target appearance. The two selection rules use the same grading backend; only the conditioning frame changes.

Table 2 Learned controls under the common local evaluator. Values are mean $\pm$ standard deviation over three training runs; the final column gives the number of runs.

Selection rule	PSNR (dB) $\uparrow$	Runs
CLIP similarity	27.0615	—
CLIP features + linear utility regressor	27.2454 $\pm$ 0.0231	3
Candidate-wise multimodal MLP	27.2375 $\pm$ 0.0191	3
Proposed set-aware selector	27.3386 $\pm$ 0.0985	3
Best candidate in pool (oracle)	29.0008	—

Table 3 Training-run stability under the common local evaluator. W/T/L counts cases that improve, tie, or regress relative to the local VCG reproduction.

Run	PSNR (dB)	Gain (dB)	W/T/L
1	27.4295	+0.6534	62/3/35
2	27.3522	+0.5761	59/6/35
3	27.2340	+0.4579	62/9/29
Mean (SD)	27.3386 $\pm$ 0.0985	0.5625 $\pm$ 0.0985	—

Table 4 Sensitivity to the number of available candidate frames.

Candidates	5	10	15	20
PSNR (dB)	27.1856	27.3063	27.3511	27.4295

4.8 Efficiency and scope

With cached image features, the selector scores one reference and twenty candidates in 4.82 ms on an NVIDIA TITAN RTX. This is selector-only latency; video decoding, feature extraction, transform estimation, and full-video rendering are separate costs. The formulation does not require a particular grading backend, but its supervision is backend-dependent: a new backend would require regenerating offline utilities while leaving the inference interface unchanged. Quantitative evidence in this paper is obtained from the VCG instantiation, and the objective target videos do not represent subjective artistic preference.

5 Conclusion

This work studied source-frame selection as part of reference-based video color-transform estimation. The central observation is that generic visual similarity does not measure whether a frame provides the structural and chromatic evidence needed by the downstream grading process. We therefore formulated the decision as relative utility learning and introduced a selector that combines structural correspondence, correspondence-guided local color matching, and set-aware candidate ranking.

On the 100-case benchmark, the complete pipeline achieves 27.43 dB PSNR versus the 24.55 dB reported for VCG, an arithmetic difference of 2.88 dB. In the matched attribution study, changing only the selection rule lowers LPIPS and CIEDE2000. Learned controls show that utility supervision is more effective than direct CLIP similarity, while candidate-set reasoning provides a complementary gain over independent candidate scoring. Stability, candidate-pool sensitivity, proxy validation, and qualitative results further characterize the decision.

The proposed selector retains a simple interface: it observes a target reference and candidate source frames and returns one frame to an unchanged grading system. Its supervision remains conditional on the downstream task, so transferring the model to another grading operator requires new offline utility labels. Future work will investigate more efficient label generation, larger and adaptive candidate pools, multiple grading backends, and human assessment of artistic preference.

Statements and Declarations

Competing interests. Yang Han and Xin Duan are affiliated with Chengdu Sobey Digital Technology Co., Ltd. The authors declare no other competing interests relevant to the content of this article.

Data and code availability. The trained selector, configuration files, evaluation records, and source code necessary to reproduce the reported selection results will be supplied in an anonymized supplementary archive at submission, subject to the benchmark licenses and the host venue’s policy. Public release will follow the same licensing constraints; benchmark data and renderer

implementations remain subject to their original distribution terms.

Author contributions. Yang Han conceived the study, developed the method, conducted the experiments, analyzed the results, and prepared the manuscript. Yaosen Chen contributed to data curation and experimental implementation. Yanru Zhang contributed to evaluation and analysis. Xin Duan contributed to project supervision and technical review. All authors reviewed and approved the final manuscript.

References

- [1] Zeng, H., Cai, J., Li, L., Cao, Z., Zhang, L.: Learning image-adaptive 3d lookup tables for high performance photo enhancement in real-time. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**(4), 2058–2073 (2022). <https://doi.org/10.1109/TPAMI.2020.3026740>
- [2] Yang, C., Jin, M., Jia, X., Xu, Y., Chen, Y.: AdaInt: Learning adaptive intervals for 3d lookup tables on real-time image enhancement. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 17501–17510 (2022). <https://doi.org/10.1109/CVPR52688.2022.01700>
- [3] Yang, C., Jin, M., Xu, Y., Zhang, R., Chen, Y., Liu, H.: SepLUT: Separable image-adaptive lookup tables for real-time image enhancement. In: *European Conference on Computer Vision*, pp. 201–217 (2022). https://doi.org/10.1007/978-3-031-19797-0_12
- [4] Zhang, F., Zeng, H., Zhang, T., Zhang, L.: CLUT-Net: Learning adaptively compressed representations of 3d luts for lightweight image enhancement. In: *ACM International Conference on Multimedia*, pp. 6493–6501 (2022). <https://doi.org/10.1145/3503161.3547879>
- [5] Chen, Y., Yang, H., Yang, Y., Liu, Y., Wang, W., Wen, X., Xie, C.: NLUT: Neural-based 3d lookup tables for video photorealistic style transfer. *arXiv preprint* (2023) [arXiv:2303.09170](https://arxiv.org/abs/2303.09170) [cs.CV]
- [6] Shin, S., Shin, D., Shin, J., Jeon, H.-G., Lee, J.-Y.: Video color grading via look-up table generation. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 19141–19152 (2025). <https://doi.org/10.1109/ICCV51701.2025.01779>
- [7] Zhang, K., Chao, W.-L., Sha, F., Grauman, K.: Video summarization with long short-term memory. In: *Computer Vision – ECCV 2016*, pp. 766–782. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-46478-7_47
- [8] Burges, C., Shaked, T., Renshaw, E., Lazier, A., Deeds, M., Hamilton, N., Hullender, G.: Learning to rank using gradient descent. In: *Proceedings of the 22nd International Conference on Machine Learning*, pp. 89–96 (2005). <https://doi.org/10.1145/1102351.1102363>
- [9] Cao, Z., Qin, T., Liu, T.-Y., Tsai, M.-F., Li, H.: Learning to rank: From pairwise approach to listwise approach. In: *Proceedings of the 24th International Conference on Machine Learning*, pp. 129–136 (2007). <https://doi.org/10.1145/1273496.1273513>
- [10] Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P.-Y., Li, S.-W., Misra, I., Rabbat, M., Sharma, V., Synnaeve, G., Xu, H., Jégou, H., Mairal, J., Labatut, P., Joulin, A., Bojanowski, P.: DINOv2: Learning robust visual features without supervision. *arXiv preprint* (2023) [arXiv:2304.07193](https://arxiv.org/abs/2304.07193) [cs.CV]
- [11] Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: *International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 139, pp. 8748–8763. PMLR, Virtual (2021). <https://arxiv.org/abs/2103.00020>
- [12] Gatys, L.A., Ecker, A.S., Bethge, M.: Image style transfer using convolutional neural networks. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp.

2414–2423 (2016). <https://doi.org/10.1109/CVPR.2016.265>

- [13] Huang, X., Belongie, S.: Arbitrary style transfer in real-time with adaptive instance normalization. In: IEEE International Conference on Computer Vision, pp. 1510–1519 (2017). <https://doi.org/10.1109/ICCV.2017.167>
- [14] Luan, F., Paris, S., Shechtman, E., Bala, K.: Deep photo style transfer. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 6997–7005 (2017). <https://doi.org/10.1109/CVPR.2017.740>
- [15] Lee, J., Lee, Y., Kim, J., Kosiorrek, A.R., Choi, S., Teh, Y.W.: Set transformer: A framework for attention-based permutation-invariant neural networks. In: International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 97, pp. 3744–3753 (2019). <https://arxiv.org/abs/1810.00825>
- [16] Chen, D., Liao, J., Yuan, L., Yu, N., Hua, G.: Coherent online video style transfer. In: IEEE International Conference on Computer Vision, pp. 1114–1123 (2017). <https://doi.org/10.1109/ICCV.2017.126>
- [17] Gao, W., Li, Y., Yin, Y., Yang, M.-H.: Fast video multi-style transfer. In: IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 3211–3219 (2020). <https://doi.org/10.1109/WACV45572.2020.9093420>
- [18] Wang, W., Yang, S., Xu, J., Liu, J.: Consistent video style transfer via relaxation and regularization. IEEE Transactions on Image Processing **29**, 9125–9139 (2020) <https://doi.org/10.1109/TIP.2020.3024018>